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The goal is to control
the national economy through computers

The State of Computation in Cuba

by Ramon C. Barquin

Computer technology has, in little over two short decades, left its imprint upon mankind. The achievements made possible by computation would undoubtedly highlight any list of postwar scientific events of major importance. Yet it hasn't been in the field of science alone that the computer has made an impact. Its capacity for information processing and data manipulation has truly revolutionized large areas of industry, commerce, banking, public utilities, services, etc. The effects of this revolution have largely contributed to the economic development process that has also characterized the more advanced nations in these last few decades. At the same time, the diversity of applications, the diminishing costs, and the ease of handling have allowed the computer to enter most of the less developed countries by now. Latin America, a developing region par excellence, saw its first computers over 15 years ago, although the existence of computation as an identifiable phenomenon does not go back beyond 1964-65. Cuba, specifically, received its first digital computer in 1959 in the form of an IBM Ramac 305.

From that beginning, although there had been prior use of punched card unit record equipment throughout the island, Cuba went through many ups and downs in its computer industry. Some of these were a consequence of the external pressures and economic vicissitudes that the country as a whole went through during the first years of the revolution. As of 1968, however, the country took a new look at the possible benefits of computation. The revolutionary leadership having been won over, Cuba embarked on a most ambitious plan to develop itself in the field of computers and their use. This plan is still under way, and it is still early to assess its results. Nonetheless, it is quite clear that in a matter of a few short years Cuba went from a point of almost zero computation, to one of rather extensive use of computers, at least within the economic planning spheres, and has become the only country in Latin America to be actively manufacturing computers of its own design. (Brazil manufactured the IBM 1401 some years back, and is now planning to build IBM 370/135s. However, this is really an IBM project in which most of the machines were actually exported. The development of the "Patinho Feo" machine at the Univerdidade do São Paulo is more in line, but it has not yet reached a definitive stage.)

The purpose of this work is basically to present the state of computer arts in Cuba today. For this the background of computation in the island will first be described, with an exposition of the history of data processing in Cuba prior to the revolution, and up through the first few years of the revolutionary regime. Following this, a presentation of the situation in Cuba today is made, emphasizing the educational program, the national policy aspects, and the decision to manufacture minicomputers. A very brief analysis of future development is also included, but is quite sketchy due to the lack of first-hand information with which to work.

As are many other areas, the Cuban experiment with computers, especially the manufacturing phase, is being observed with extreme interest by most Latin American countries. If Cuba can prove that its minicomputers are technologically competitive and economically sound, then it will probably become a model for a number of nations with similar needs. If it cannot—and the critical test will be whether it can solve the majority of its computational needs over an extended period, and at a lower cost than that of importing the computers—it will at least have determined the advisability for a developing country to manufacture its own computers.

Lastly, it should be noted that for an adequate treatment of the subject, first-hand research is necessary. The author has had to rely on the scant written material that exists, and interviews with a few computer-related persons who have visited Cuba, or worked there within the last few years.[1]

In 1927 the United Railways of Havana (Ferrocarriles Unidos) brought the first adp equipment to Cuba. This was in the form of IBM punched card machines which were to be used in administrative applications for the railroad company. The success of this innovation soon made the Consolidated Railways of Cuba (Ferrocarriles Consolidados) also adopt the use of IBM machines, and in 1928, shortly after their installation, the IBM Company appointed its first representative to Cuba in the person of John A. Griffin, who would later on become the head of IBM's main office for Northern Latin America. By 1931 a number of other enterprises, including the City of Havana, were using the punched card machines, and the 1930 census had been processed using adp equipment. In fact, business for IBM was sufficiently good for the establishment of a permanent branch office in Havana in 1933. This office, a branch of the IBM Co. of Delaware, had Chester Lecher as its first manager, and was also responsible for Puerto Rico, Panama Canal Zone, Nicaragua, Jamaica, Haiti, Costa Rica, and the Dominican Republic. Chester was replaced by R. B. Guijarro in 1935, and later on by Marcial Digat in 1940. Digat remained as IBM country manager in Cuba until the confiscation of IBM by the Castro government in 1961.

Not all adp equipment that entered the island prior to the revolution was IBM manufactured; there were some Sperry Rand and NCR machines also, but in almost negligible numbers. The history of dp in Cuba during this phase is basically the history of IBM in Cuba. There were numerous unit record installations throughout the country. Most of them were within the private sector of the economy. The public sector had not advanced very much in the use of dp equipment. The majority of the machines were concentrated in Havana, although the diffusion of demand was great enough to warrant the establishment of a new IBM office in Santiago de Cuba by 1954. Towards the latter years of this phase, and thus the situation just prior to the revolution, the types of equipment to be found in the island were typically keypunches and verifiers, sorters, collators, reproducers, interpreters, and accounting machines. These last were

[1] Most of the information presented here comes from the following: J. Connolly, *A Chronology of Computing in Europe, Africa, Asia and Latin America*, IBM World Trade Corp., N.Y., 1968; O. Carnota Lauzán, *The Use of Computers in the Economic and Social Field in a Developing Country: Cuba*, Conference on the Role of Computers in Economic and Social Research in Latin America, Cuernavaca, Mexico, October 25-29, 1971; and Interviews with former employees of IBM de Cuba.

probably of the IBM 402, 403, and 407 types. The first IBM 632 machine to be installed in Latin America was the one at Incera y Hnos. of Havana. There were also some calculators of the IBM 602 variety.

By 1959 there was a rather progressive IBM organization in Cuba that had managed to establish a firm base for commercial dp in the country, placing it in a privileged position technologically, with respect to most of the other Latin American countries. In effect, by this time both the country and the local IBM organization were ready to bring the first digital computer into Cuba. This was to be a Ramac 305, to be installed at IBM's office in the La Rampa building in Havana, and later to be passed on to the Cuban government after some final commercial agreements had been reached. By the time all of the devices had been received in Cuba, and an attempt to install them had been made, the years 1960 and early 1961 had come around. With them came many of the major confrontations of the early revolution; the migration of most of the IBM technicians, and a general lack of interest on the part of the government to devote a great deal of its resources to the endeavor. The Ramac 305 apparently did become operational in 1961, but that was about all that could be said. It does not seem probable that any useful work was ever obtained from it. At least this was the case up to the time of the confiscation of IBM's equipment in Cuba. (Apparently the computer was actually installed by an IBM customer engineer who was promised that he could leave the country if he got it working. This he did, but once the computer needed maintenance for the first time, there was probably no one to provide it.)

Some comments have been made concerning the actual arrival at the docks in Havana of some computers which were returned to the U.S. by IBM before they were allowed to be delivered to the customers in Cuba during the 1959-60 period. This whole issue is not very clear, and Carnota mentions some punched card machines that were re-embarked. However, it is also possible that an IBM 1401 which had been ordered by the Cuban Telephone Co. and scheduled for arrival about that time could have been reshipped due to the rapidly deteriorating political situation in the country. In any case, it is clear that prior to the revolution there was no digital computer in operation inside of Cuba, and that the first two years saw only the arrival of the Ramac 305 which was never really usable.

Data processing had reached a point of maturity in the country, however, and most of the major commercial schools in the nation were establishing curricula. Thus one could see keypunching, basic machine operations, wiring, and mechanized accounting being taught on unit record equipment at Colegio Baldor, Academia Pitman, Havana Business School, and the Universidad Masónica, among others.

The confiscation of IBM was an interesting process, since there were no major industrial facilities (except a card plant) to actually take over. In 1960 the IBM manager, Marcial Digat, was called back to New York in view of the Castro government's initial confiscations of U.S. companies. Shortly afterwards, Nicanor Infiesta was named to represent IBM interests in Cuba, and in 1961 the representation fell on IBM's general accountant, Manuel Rodríguez. Further on during that year the government decided to confiscate IBM, thus acquiring the stock of unit record punched card machines installed in the country up until that time. It also placed them in the position of having to obtain spare parts to maintain the machines without recourse to the IBM Co. directly. This they managed to do indirectly for some time, through rather ingenious methods. The major blow to the country as a whole, as far as dp and subsequently computation were concerned, was the emigration of many technicians, operators, programmers, analysts and managers. Almost all of IBM's Cuban staff left the land and went on to provide the IBM Latin American organization with a large number of country managers for Central and South America, and to strengthen the technical staffs of these countries. Cuba's loss was their gain, and still today one finds IBM's offices in these countries dotted with Cuban technicians who left after the revolution.

Some early attempts to establish a good educational base for computation and to implement sound programs apparently came from Mexico in the 1960-61 period, and later on from the East bloc countries, especially Czechoslovakia.[2] None of these attempts seem to have been very successful, probably because their priority was not sufficiently high within the overall scope of the central leadership. In addition, a scarcity of trained personnel, or of personnel with the necessary background, must have been in evidence. Up to and including 1967, Cuba was only able to acquire one computer, an Elliot 803 which was bought from Great Britain without peripherals. Carnota seems to suggest that they could not get peripherals with the machine for political reasons.

The year 1968 seems to mark the dividing line between the old and the new in Cuban dp or computation. If 1959 marked the coming to power of the revolution, 1968 was the year of the revolution in computation.

A need for computers

In 1968, apparently through the agencies of French advisors, the top leadership was convinced of the need for using computers for the solution of many of the major problems confronting the Cuban economy, primarily in order to plan properly. As a result of this decision a number of events took place in rapid succession. First, two SEA 4000 second generation computers were ordered from CII (Compagnie Internationale pour l'Informatique), and the Dirección de Cálculo Electrónico (Central Directorate for Electronic Calculation) was created under the Central Planning Board (JUCEPLAN). Next, in 1969, the Centro de Investigacion Digital (CID) was founded at Universidad de La Habana to study the problem of manufacturing minicomputers in Cuba. And in that same

Computer CID-201-A	
Number	**Location**
2	CID
1	Central Azucarera Camilo Cienfuegos
3	Ministerio del Azucar
1	CEIS
1	Escuela de Matematicas
1	Instituto Cubano del Petroleo
1	Industria Textil Ariguanabo
1	Industria Nacional Deporte y Recreacion
1	Ferrocarriles
1	Central Azucarera Ecuador
1	Construimport
1	Maguimport
1	MINAZ
1	Grupo Nacional de Genetica
1	Instituto de Economia
1	Mineria

Source: Report of Prof. Jose Duran, from Universidad de Concepcion, Chile, on his visit to Universidad de La Habana, Cuba, 1972.

Table 1

[2] Personal communication to the author by Ing. Sergio Beltrán, who was then director of the Centro de Cómputos at the Universidad Nacional Autónoma de México; and from Mr. Al Strnad, systems analyst for the Czechoslovak govt., later a research associate at MIT.

year the Plan Cálculo Nacional, modeled on the French Plan Cálcul, was established. Lastly, the preparation of study programs for the rapid training of personnel at the different levels was begun.

All of this feverish activity, which took place because of the sudden importance attributed to computation by the revolutionary leadership, gave the island a major set of goals to achieve. These can be divided into four major categories: education, applications, manufacturing, and planning.

Insofar as the need existed to train new personnel for the computer industry, a number of plans were formed to satisfy this need. At first the French and other foreign instructors were used for the first set of courses at the University of Havana, and subsequently Cuban personnel took over the teaching. There was a great addition of computer-related courses within the formal engineering curricula at the universities, as well as in those for mathematicians, physicists, economists, and public accountants. A degree program in systems engineering and a graduate program at the MS level were begun in 1972. The Universidad de La Habana and the Universidad Central de Las Villas also offer the degree of Licenciado en Ciencias de Computación under the Mathematics curriculum. Lastly, the Technical Institutes in Economy turn out the mass of computer programmers that are needed, while the Technical Institutes in Electronics produce the hardware maintenance technicians.

As far as applications go, the government seems to have undertaken the automation of a national cattle control system, a population and housing census, a sugar crop control system, etc. There have been major limitations as to the implementation of these systems due to the lack of hardware, but they should be working fairly well in theory by now. There are other applications which demonstrate the Cuban government's preoccupation with prestige issues. The most important is massive sports archives and statistics. Cuba's chances of winning a medal at an international event can be assessed in this manner, and the decision made whether or not to send a representative.

Creation of a mini

Probably the most important aspect of Cuba's post-1968 computation programs has been its decision to manufacture minicomputers. Upon the creation of the CID with five engineers and a mathematician, the job was assigned them of studying the feasibility of manufacturing computers in Cuba. The main justification found for this was the huge payoff to be obtained by the optimization of rail traffic during the sugar harvest. Just this application, with the gigantic 10 million ton prospective harvest of 1970 in mind, drove the full staff at the center to complete the first computer in April 1970. After slight modifications the final version, called the CID-201-A, became operational. This machine, which may be classified as a minicomputer, is a 4K, 12-bit word, 1.5-usec cycle machine. The CID-201-A is programmed using the LEN I assembler, or the LEAL compiler. LEAL is a high-level language similar to Digital Equipment's FOCAL. (The PDP-8 is a 4K minicomputer quite similar to the CID-201-A. Prof. Carlos Domingo, of Universidad Central de Caracas, Venezuela, who spent a year as UNESCO advisor on computation in Cuba, suggests the possibility of Cuba's having borrowed from this original design through Digital's very fine set of manuals.) The computer is a bit limited in its peripheral equipment, however, having a poor assortment of I/O devices. At present there is only a teletype-printed output at 7 cps. Paper tape input and output is also standard, as is magnetic tape cassette with 35 word/second read-write speed.

In July of 1971 the CID-202 was developed, with basically the same characteristics, except for a 16-bit word and 8K memory. Work was under way on a CID-201-B, whose specifications are not yet known.

Programming the CID machines seems to be fairly straightforward with the use of the two languages and the program library. Lenguaje Ensamblador I (LEN I) is an assembler language that permits the use of mnemonics to substitute for the actual machine code. Lenguaje Algorítmico (LEAL) is a compiler which permits the writing of code in a higher level fashion. It is similar to DEC's FOCAL, which is used on the PDP-8. LEAL was developed at CID by Victor M. Llopis and collaborators.

The program library includes most of the I/O subroutines and the systems utilities as well as some of the mathematical ones. The list as presented in the specifications manual is:

1. Paper tape read
2. Paper tape punch
3. Paper tape verification
4. Decimal format keyboard input
5. Decimal format keyboard verification
6. Floating point addition
7. Floating point subtraction
8. Floating point multiplication
9. Floating point division
10. Transcendental functions
11. Monitor and trace programs
12. Core dump routines

Software development is really the main work ahead of the CID group at present. The hardware works, apparently, so now it must be put to good use, and here is where software is central. One aid to this effort seems to be a mutual deal whereby three CID machines were given to Chilean universities in exchange for software development.

At present, the CID seems to have built around 40 computers. Of these, approximately 20 are listed in Table 1 with their respective installation sites. Table 2 provides the characteristics of both the CID-201-A and the CID-202.

The planning function seems to be extremely important in Cuba today, at least in theory. Designing a computer system for planning is thus also quite important. The Cuban government has undertaken the implementation of a national computing system (Sistema Nacional de Computación) which is to be the fundamental instrument for controlling the Cuban economy. The grandiose master plan envisions a national switching and data processing center as the main depository, with a telecommunications network and decentralized systems providing the whole country with access to the data. Although this system is still far from implementation, the first steps have been very deliberately taken to bring it

Characteristics of the CID-201-A and the CID-202

Specification	CID-201-A	CID-202
Memory type	magnetic core	magnetic core
Memory capacity	4,096 words	8,192 words
Cycle time	1.5 usec.	1.5 usec.
Word length	12 bits	16 bits
Instruction set	59 inst.	200 inst.
Additions per second	25,000	50,000
Peripherals:		
Teletype	yes	yes
Paper tape read	yes	yes
Paper tape punch	yes	yes
Cassette recorder	yes	yes
Auxiliary storage capacity	240K words	240K words
Multiprogramming	no	yes
LEN I (Assembler language)	yes	yes
LEAL (Algorithmic language)	yes	yes
Development date	April 1970	July 1971

Source: "Computadora CID 201-A," Centro de Investigacion Digital, Universidad de La Habana, Cuba, 1972, and "Computadora CID 202," Centro de Investigacion Digital, Universidad de La Habana, Cuba, 1972.

Table 2

about. The equipment to be used has been ordered from CII, and is probably in the process of being installed in Cuba now. Carnota lists it in some detail being composed primarily of IRIS series machines: 2 IRIS 50's and 9 IRIS 10's. The IRIS computers are third generation equipment, and have data communications facilities. They will be working on 1200- and 2400-baud voice grade lines.

The final objective

Cuba seems to have developed a very definite strategy, at a national level, with respect to computation. This strategy covers both application goals and allocation of computer resources. It has as its final objective the establishment of a national computation system which will control the national economy. This system will span the island through the use of data communications facilities and several intercommunicating IRIS series computers. The allocation of computer resources seems to follow a two-part path. First, importation of some medium to large third generation machines with which to implement the NCS. Secondly, but simultaneously, the design and manufacture of small digital computers to satisfy the problems which characterize typical large enterprise operations such as sugar mills, large factories, large farms, etc.

The NCS idea is not new, and socialist regimes, because of their central planning and control, usually identify with the scheme quite strongly. There are many pitfalls that could conceivably hamper the actual operation and effectiveness of such a system. One of the most evident problems is the reliability and timeliness of the information collected. This is the base of any information system, and great care must be taken to ensure that the data gathered meets both requirements. Another possible difficulty in the implementation of such a system is the complexity problem. The software for the NCS will have to be written quite for that specific purpose. The magnitude of the project being considered bears the characteristics of a major systems development undertaking. If the experience of almost everyone who has developed a major system holds, there will be a tendency to grossly underestimate the magnitude of the problem. This is even more likely to occur in Cuba because of the lack of experience in software development and major systems projects. The intent of the project is definitely worth a good try, and Cuba seems determined to give one.

The establishment of a national computation system with a universal data base encompassing all sorts and types of economic information could have terrifying effects on individual privacy. Yet, because of the very nature of the present Cuban regime, individual privacy has very low priority, and the centralized authoritarian regime which has characterized the revolution would probably prize some additional elements of control.

By deciding to manufacture its own computers, Cuba departed from the accepted practice among most developing countries of importing its dp equipment. Usually economies of scale, the cost of R&D, and the impossibility of competing with the major manufacturers have made this appear to be a sound policy. Cuba was impeded, for political reasons, from purchasing adequate computing equipment in the U.S., and what it could get from the East bloc was not satisfactory technologically speaking. The French seemed eager and capable of satisfying certain requirements concerning hardware by late 1967. But with the creation of the CID, and some additional advice received from the start, Cuba decided to build its own computers at the lower end of the scale. They would import what they needed for immediate operations, along with some of the large machines which they could not conceivably justify constructing. The CID would develop minicomputers with which to solve the basic problems of medium to large scale Cuban firms. Was the decision correct? Should it be followed in other developing countries? It is still too early to reach a conclusion on either inquiry. Certainly the fact that Cuba was not able to freely acquire U.S. computers places it in a unique position which from the start casts doubt on the further applicability of the model. At the same time, it becomes clear that it could probably be justified in the Cuban case due to this special circumstance. But in order to reach a better conclusion on the questions asked, certain facts should be collected. The costs of building the CID machines—how are they broken down? How much is allocated to materials? Where does Cuba obtain the components? Does she manufacture them also? What were the real R&D costs? What other projects could have been accomplished by the same people at the same cost? What is the life cycle of the CID-201-A? Of the CID-202? How long before it becomes obsolete? What is the cost of software development vs. use of a system with ready-to-use software? These and many other questions need answers before a final evaluation can be made concerning the Cuban experiment. There is no doubt, however, that it is being closely watched and will probably be emulated before long. This is another point which should be recognized when analyzing Cuba's decision to build computers. By undertaking this project, Cuba highlights her technological prowess, thereby gaining considerable prestige, especially among the developing countries, which is central to the conduct of its foreign policy.

Lastly, in order to place Cuba comparatively within the context of Latin America, it is necessary to say that it has a CIDP ranking of 12th, and belongs in Group C of the corresponding categorization of the Latin American nations.[3] Other members of this group include Chile, Colombia, Uruguay, Peru, Panama, and Costa Rica. Cuba, with a CIDP of 31.63, is towards the lower end. Because of the peculiar conditions governing the island's development within the computer industry, the CIDP concept does not seem to be specifically applicable. Nonetheless, it gives an idea of the general position. It also corresponds approximately to the basic level as defined by the United Nations ad hoc Committee on Computer Technology.[4] In addition, Cuba has 4.85 computers per million inhabitants, which is below the median for the region and compares with Chile's 5.88 and Colombia's 4.07. It also has 17.59 computers per billion dollars of GNP. This is again below the region's median, but above Chile's 12.54 and Colombia's 16.73. □



Mr. Barquin is presently on a leave of absence from IBM to complete his PhD at MIT. He has previously been an instructor and a systems engineer at IBM in San Juan, and has BS and MS degrees from the Univ. of Puerto Rico. He was born in Cuba, and came to the U.S. in 1960.

[3] The Computer Industry Development Potential (CIDP) index is a measure of the short term potential of a country for developing a computer industry. It is computed through the weighing of 11 variables of economic, educational and technological character. See R. C. Barquin, *Development of the Computer Industry in the Latin American Nations: Strategies and Policy*, Electrical Engineer's Thesis, MIT, Cambridge, Mass., 1972.

[4] *The Application of Computer Technology for Development*, United Nations publication, NYC, 1971.